Chittagong University of Engineering & Technology
Dept. of Computer Science & Engineering

CUET Competitive Programmer's Syllabus

A guideline for the students of CUET for
'Competitive Programming'

CUET Computer Club

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CUET Competitive Programmer's Syllabus

About:

This is a guideline for the Competitive Programmers of CUET in accordance with their academic calendar. It contains the topics that must be covered within first two years if you want to qualify for the world finals. The syllabus is divided into four terms. Each term covers some topic containing detailed tutorials and problem links. Level-1 Term-1 covers the basics of programming Language. Level-1 Term-2 is the most crucial part since it introduces the basic idea of data structures/algorithms. In Level-2, intermediate/some advanced topics are covered. The syllabus of Level-3 and Level-4 is not provided here. It's your own duty to find out your lacking and focus/specialize on them. The links provided here will help you in a great deal but there might be many more resources. So always keep yourself up-to-date.

Goal:

To see CUETians in ICPC World Finals. ☺

Shortcuts:

1. Level-1 Term-1
2. Level-1 Term-2
3. Level-2 Term-1
4. Level-2 Term-2
5. Level-3 & Level-4

Some Important Links:

1. Code Templates
2. Resources in Bengali
3. Books Related to Competitive Programming

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Term wise Syllabus

Level-1 Term-1

Tentative Class Schedule:

Week	Topic
1	Variables, Data Types, Scanf/Printf, Format Specifier
2	If-else, Nested If-else
3	Loop Basics: While, Do-while, For-loop
4	Nested Loop, Break, Continue
5	Loop Advanced: Loop+If-else
Online Contest-1	
6	Array: 1D, 2D
7	Character Array, String
8	Function, Recursion
9	Binary Search + Others
10	STL Basic
Online Contest-2	

Guidelines:

- Get complete idea about C
- Get basic idea about C++
- Try to implement what you think
- Must participate in Long-Contests (Upto 5 days) that will be held after each class
- Take help from your Batch-mates, Seniors and Teachers if needed

Expectations:

- Able to implement your idea
- Familiarize with all online judges
- Read Tamim Shahriar Subeen's Book completely
- Solve all the problems of LightOJ beginner's category
- Total Number of problems solved: 200+
- CF rating:
  - -At the start: 900+
  - -At the end: 1200+

Tips:

- Learn about different good resources as well. Geeksforgeeks, emaxx, competitive programming 3, Mahbubul Hasan Shanto vai's book on programming contest, Shafaet Ashraf vai's blog. These are good, but not just these-there are others. Google is your friend.
- Spend around an hour or two at each problem before searching for solutions online. Not more than that, because what is important for you at this stage is to solve more and more problems, and if you spend all your day at one problem, your overall solve count will be low and you will not learn much. But make sure that you spend this time wisely-when we say we have worked for an hour, what happens in reality is that we have worked for just 20 minutes in total, and the other 40 minutes have been wasted away via other 'useful' procrastination. Perhaps learn to adopt the Pomodoro method.
- After seeing the solution to a problem, make sure to understand the solution fully and absolutely. There should be no doubt regarding anything with the solution, and if there is, that means you have not understood it fully/internalized it. Problem solving is more about pattern matching, and being able to match an unseen problem with problems that you have seen before-but if you haven't internalized the solution to the problems that you couldn't solve by yourself, you have then actually just memorized it, and memorized stuff doesn't actually stay in your mind for long-and you won't be able to recognize patterns later as well.

Tutorials & Problems List for Level-1 Term-1

(1-1) Week 1 (Variables, Data types, Scanf/Printf, Format Specifier)

Tutorials:

1. [প্রোগ্রামিং বইঃ অধ্যায় দুই] ডাটা টাইপ, ইনপুট ও আউটপুট।
2. List of all format specifiers in C programming
3. Data Types in C
4. https://youtu.be/e9Eds2Rc_x8

Problems:

1. Codeforces: 1A, 99999100
2. Atcoder: ABC153A, ABC-148A
3. Hackerrank: Playing with Characters
4. URI: 1001,1002,1006,1015,1019

(1-1) Week 2 (If else, Nested If else)

Tutorials:

1. Decision Making in C / C++ (if , if..else, Nested if, if-else-if)
2. কম্পিউটার প্রোগ্রামিং বই: [প্রোগ্রামিং বইঃ অধ্যায় তিন] কন্ডিশনাল লজিক।

Problems:

1. Codeforces: 122A, 263A
2. Codechef: LEAP, EO, ASET002
3. Atcoder: ABC152A, ABC065A
4. URI: 2582
5. UVA: 11172
6. Hackerrank: Conditional Statements in C

(1-1) Week 3 (Loop Basics: While, Do-While, For-loop)

Tutorials:

1. [প্রোগ্রামিং বইঃ অধ্যায় চার] লুপ (Loop)।
2. Loops in C and C++

Problems:

1. DimikOJ: Problem 6, Problem 3, Problem 5
2. Timus: 1149
3. LightOJ: 1000, 1022, 1216
4. Codeforces: 99999123, 791A, 977A

(1-1) Week 4 (Nested Loop, Break, Continue)

Tutorials:

1. Nested Loops in C with Examples
2. Difference between continue and break statements in C++

Problems:

1. CodeChef: BICBPAT1, PPATTERN
2. URI: 1189, 1190, 1183, 1435, 1478, 1186, 1188
3. Hackerrank: Printing Pattern using Loops

(1-1) Week 5 (Loop Advanced: Loop + If-else)

Tutorials:

1. [প্রোগ্রামিং বইঃ অধ্যায় চার] লুপ (Loop) I

Problems:

1. Codechef: LEAPY
2. Atcoder: 079A
3. Timus: 1083, 1209
4. LightOJ: 1001, 1015, 1053, 1008
5. URI: 1557, 1187

(1-1) Week 6 (Array: 1D, 2D)

Tutorials:

1. অ্যারে - Array Archives - হাসানের রাফখাতা
2. Fred: C++ Notes: Table of Contents (See the Arrays section)
3. অ্যারে কম্প্রেশন
4. 1-D Tutorials & Notes | Data Structures (try to solve some problem from this problem section)
5. How to visualize multidimensional arrays
6. Array Data Structure
7. Lecture Notes on Arrays

Problems:

1. UVA: 10038 (Jolly Jumpers), 414 (Machined Surfaces), 665 (False coin), 467 (Synching Signals), 10703 (Free spots), 10855 (Rotated square), 10920 (Spiral Tap),11349 (Symmetric Matrix), 11581 (Grid Successors), 12187 (Brothers)
2. Codeforces: 1213B, 1265B, 1279C

3. Codechef: SALARY

(1-1) Week 7 (Character Array, String)

Tutorials:

1. Strings in C
2. Strings in C (With Examples)
3. C++ Strings
4. string - C++ Reference
5. C++ Strings

Problems:

1. Codeforces: 71A, 1146A, 118A, 1139A, 1223B, 1243B1, 448B, 1151A, 165C, 1243B2, 1268A, 1153C

(1-1) Week 8 (Function, Recursion)

Tutorials:

1. রিকার্শন (Recursion)
2. recursion Archives - হাসানের রাফখাতা
3. recursion_and_dp - smilitude
4. Recursion and Backtracking Tutorials & Notes | Basic Programming

Problems:

1. UVA: 624, 167, 539, 729, 750, 628, 10276, 11085, 574, 524, 10063, 10503, 301, 331, 193, 129, 208, 416, 861, 1262

(1-1) Week 9 (Binary Search + Others)

Tutorials:

1. Binary search implementation (Implementation variations)
2. Binary Search
3. Binary Search Practice Problems | Algorithms | page 1
4. বাইনারি সার্চ অ্যালগরিদম - Binary Search Algorithm
5. Ternary Search Tutorials & Notes | Algorithms
6. Ternary Search and its Applications
7. The great ternary search hoax

Problems:

1. Codeforces: 492B, 474B, 195B, 1221C, 1183C, 371C, 812C, 706B, 580B, 340D, 75C, 448D, 1250J, 616D, 1077D

2. LightOJ: 1043, 1072, 1307, 1138, 1088,1048, 1137, 1235,1383, 1146, 1240
3. Spoj: AGGRCOW, ACTIV
4. Codechef: MCO16503, KGP16G

(1-1) Week 10 (STL Basic)

Tutorials:

1. Power up C++ with the Standard Template Library: Part I
2. Power up C++ with the Standard Template Library: Part II: Advanced Uses
3. stl - smilitude
4. স্ট্যান্ডার্ড টেম্প্লেট লাইব্রেরী
5. C++ STL (Bangla)
6. STL

Problems:

1. UVA: 146, 10107, 514, 10935, 484, 10815, 10954, 400, 10194, 127, 540, 1203, 12058, 855, 11321, 10264, 732, 1062, 10172, 978

Level-1 Term-2

Tentative Class Schedule:

Week	Topic
1	Basic Math, Modular arithmetic, Big-mod, Modular inverse
2	Number Theory: Sieve, Prime Factorization, Number of divisors
3	Recursion, Backtracking, Tail-call recursion
4	Greedy Techniques, Task Scheduling, Kadani's Algorithm
5	Basic DP (Recursive/Iterative dp, knapsack, LIS)
Online Contest-1	
6	Advanced STL
7	Graph Basics: BFS/DFS
8	Shortest Path: Dijkstra, Floyd-Warshal, Bellman-Ford
9	Topological Sort, Tree-Diameter
10	Basic Geometry
Online Contest-2	

Guidelines:

- Participate in online contest
- Upsolve problems after attending online contest
- Try to attend and do well in onsite contest

Expectations:

- Total Number of problems solved: 300+
- CF rating:
  - -At the start: 1200+
  - -At the end: 1500+

Tips:

- Learning how to think is important at this stage. You should have gotten a feel for determining how much time you should give to each problem in order to learn properly-however, we would say do not make it less than 30 minutes, and definitely not more than 90 minutes. But make sure that you have utilized that time of thinking fully-many of us do not utilize this time fully and do not realize it. Learn to think in a IDA* approach, that is, realizing all problems have an easy solution, and can be solved in a few steps. If you find your train of thought getting too complex, or taking too many steps/difficult to visualize, then it is likely not correct, and if it is, it is most definitely not the intended solution. Learn to abandon current train of thought as soon as you realize that it may be getting too complex for you-these problems are made for humans by humans, not for Einsteins by Einsteins.
- You should have also realized that this **competitive programming business is no joke**. The future world finalists are currently working very hard, and while you are watching movies/browsing the net, they are currently solving problems at their ACM Room. You should spend most of times on thinking about problems. You can also think about problems when eating, bathing or even before sleeping
- Also, do not pick too hard problems, problems in the difficulty range where you find yourself seeking the solution almost every time. Even if you feel that you have understood the solution fully, you may not have fully internalized it-and possibly even indirectly memorize it. Ideally you must pick problems, that, if they were to appear in a contest, you can almost get the idea, but couldn't get a clear picture of the full solution within contest time.
- And at the minimum, **do every CF contests. Even if there is a CT the next day**. Solve the problems that you couldn't solve the next day. This is very, very, very important. This is called upsolving, and upsolve the next one or two problems that you couldn't solve in contest time. Perhaps leave the others for now.
- Please do look at the official solution (editorials) of problems, even if you have managed to solve one by yourself. You may find new ideas/trick there. Make a habit of learning from other people's code too.
- Look at other people's code to learn new tips/tricks after solving a problem.
- And do hide CF tags. 50% of solving a problem is about identifying the correct category of the problem, and you'll be stunted in this respect.

Expected Number of Problems solved (including previous semester)- 500 at the very least, more like 1000 if you hope for qualifying for WF in the future.

Tutorials & Problems List for Level-1 Term-2

(1-2) Week 1 (Basic Math, Modular arithmetic, Big-mod, Modular inverse)

Tutorials: (Basic Math)

1. Understanding class 1-12 math is enough
2. Euclidean Algorithm - Greatest Common Divisor
3. Lowest Common Multiple of Two Number

Problems: (Basic Math)

1. LightOJ: Basic math category
2. Codeforces: Math tag ordered by rating

Tutorials: (Modular Arithmetic & Big-Mod)

1. মডুলার অ্যারিথমেটিক
2. Introduction to Modular Arithmetic
3. Powerful tricks with calculation modulo

Problems: (Modular Arithmetic & Big-Mod)

1. LightOJ: 1067, 1054, 1102, 1419
2. UVA: 374, 10127, 128, 10176, 10212, 10489, 11029

Tutorials: (Modular Inverse)

1. Modular Multiplicative Inverse
2. Modular Inverse

Problems: (Modular Inverse)

1. UVA: 11904
2. CF: 300C, 622F, 717A, 896D

(1-2) Week 2 (Number Theory: Sieve, Prime Factorization, Number of divisors)

Tutorials: (Number Theory)

1. Category Archives: Number Theory
2. Sieve of Eratosthenes How Many Divisors of a Number
3. Generate Divisors of N (Move to 2-1)
4. A_Bunch_of_Stuff_:_Number_Theory_in_Competitive_Programming_[Tutorial] (Optional)

Tutorials: (Sieve)

1. প্রাইম জেনারেটর (Sieve of Eratosthenes)
2. Sieve of Eratosthenes - Generating Primes
3. Sieve of Eratosthenes

Problems: (Sieve)

1. LightOJ: 1035, 1028

Tutorials: (Bitwise Sieve)

1. বিটওয়াইজ সিভ(Bitwise sieve)

Problems: (Bitwise Sieve)

1. SPOJ: TDPRIMES

Tutorials: (Segmented Sieve)

2. Segmented Sieve of Eratosthenes
3. Book: Programming Contest, Data Structure and Algorithm – Mahbubul Hasan Shanto (segmented sieve topic, page-55)

Problems: (Segmented Sieve)

1. SPOJ: PRIME1, PAGAIN
2. LightOJ: 1197

Tutorials: (Prime Factorization)

1. Prime Factorization of an Integer

Problems: (Prime Factorization)

1. UVA:
    - EASY: 583, 160, 993, 10299, 11466
    - MEDIUM: 10392, 10179, 10139, 10484
    - HARD: 10622, 10680, 11347

Tutorials: (Number of Divisors)

1. Number of Divisors of an Integer

Problems: (Number of Divisiors)

1. SPOJ: COMDIV

(1-2) Week 3 (Recursion, Backtracking, Tail-call recursion)

Tutorials:

1. Attacking Recursions
2. টেইল-কল রিকার্শন অপটিমাইজেশন
3. রিকার্সিভ ফাংশনের সৌন্দর্য - ১
4. https://youtu.be/J8mChdOJWOs

Problems:

1. UVa: 624, 750, 10576, 11085, 524, 574, 10503, 193, 416, 10094

(1-2) Week 4 (Greedy Techniques, Task Scheduling, Kadani's Algorithm)

Tutorials: (Greedy)

1. Greedy is Good
2. Basics of Greedy Algorithms Tutorials & Notes | Algorithms
3. Greedy Algorithms

Problems: (Greedy)

1. BAPS Greedy and Searching Problem List
2. Easy:
    - CF - 479C
    - SPOJ – GERGOVIA
    - CF - 58B
    - UVA – 410
    - POJ - 1328
3. Medium:
    - SPOJ – MSCHED
    - CF - 313C
    - SPOJ – CMIYC
    - SPOJ – DRAGON
    - CF - 166C
    - Hackerrank - Goodland Electricity
4. Hard:
    - CF - 316A1
    - Codechef - Protecting The Poison
    - CF - 794C
    - SPOJ – TROOPS
    - A2OJ-306

Tutorials: (Task Scheduling)

1. Activity Selection Problem | Greedy Algo-1
2. Weighted Job Scheduling

3. Job Sequencing Problem
4. Page Replacement Algorithms in Operating Systems
5. Activity Selection Problem

Problems: (Task Scheduling)

1. Easy:
    - CSES - 1630
    - CSES - 1629
    - CSES - 1619
2. Medium:
    - CSES - 1164
    - Leetcode - 452
    - SPOJ - MIA14B
3. Hard:
    - CF(gym) - 101498F

Tutorials: (Kadane's Algorithm)

1. Search the subarray with the maximum/minimum sum
2. Kadane's Algorithm — (Dynamic Programming) — How and Why does it Work?
3. Largest Sum Contiguous Subarray

Problems: (Kadane's Algorithm)

1. Easy:
    - CSES – 1643
    - UVA – 507
    - UVA – 10684
    - Timus - 1146
2. Medium :
    - UVA – 787
    - CodeForces - 75D

(1-2) Week 5 (Basic DP-Recursive/Iterative dp, knapsack, LIS)

Tutorials:

1. ডাইনামিক প্রোগ্রামিং – ১ (ফিবোনাচ্চি)
2. https://www.youtube.com/playlist?list=PLrmLmBdmIlpsHaNTPP_jHHDx_os9ItYXr&app=desktop
3. Dynamic Programming: From Novice to Advanced
4. Tutorial for Dynamic Programming

5. https://www.youtube.com/playlist?list=PLl0KD3g-oDOGJUdmhFk19LaPgrfmAGQfo&app=desktop

Problems:

1. UVa - 787, 10684, 108, 11951, 10827, 111, 481, 11456, 10130, 10616, 674
2. CF - DP tag

(1-2) Week 6 (Advanced STL)

Tutorials:

1. https://ishtiaqhimu.blogspot.com/2020/10/standard-template-library-stl.html
2. Containers - C++ Reference

Problems:

1. UVa - 514, 732, 1062, 10172, 10901, 11034, 11572, 11308, 978, 11849, 1203, 11995
2. SPOJ - CTRICK
3. Timus - 1521

(1-2) Week 7 (Graph Basics: BFS/DFS)

Tutorials:

1. গ্রাফ থিওরিতে হাতেখড়ি - ১ | শাফায়েতের ব্লগ
2. Graph Editor (for graph drawing)
3. Depth First Search (DFS) (Silver) (***)
4. Breadth First Search (BFS) (Gold) (***)
5. me-shaon/bangla-programming-resources: Bangla tutorial, reference and resource list on programming topics (graph theory portion)
6. Programming Camp 2020 Session 3: Graph Theory
7. https://mehedi6022.blogspot.com/2019/02/breadth-first-search.html (BFS)
8. https://mehedi6022.blogspot.com/2020/02/0-1-bfs_11.html (0/1 bfs)
9. POTW (see Graph Theory I)
10. Video Tutorial (BAPS): https://www.youtube.com/watch?v=xswqISXxAC4&list=PLWtSipmftM8pGx7rR9xviLokw29kG_s-2&index=9
11. (Code): https://www.youtube.com/watch?v=VW85xQ6GJP4&list=PL2q4fbVm1Ik6DCzm9XZJbNwyHtHGclcEh

Problems:

1. Easy

- BFS/DFS Category(LightOJ)
- Building Roads
- USACO
- Message Route

2. Medium:
    - USACO (Flood Fill)
    - USACO
    - CSES-1668(Bicoloring), 1682 (Directed dfs)
    - USACO
    - Counting Rooms (Flood Fill)
    - Icy Perimeter (Flood Fill)
    - Where's BESSIE?(Flood Fill)
    - Why Did the Cow Cross the Road III (Flood Fill)
    - Labyrinth
    - Tree Diameter
    - Subordinates (DFS on Tree)
    - Codeforces: 862B,
    - https://csacademy.com/contest/archive/task/bfs-dfs
    - DCP-60: Holloween Party Back to All Problems (0/1 bfs)
    - Building Teams
    - Round Trip
    - Monsters
    - SPOJ.com - Problem AKBAR
3. Hard:
    - USACO
    - USACO
4. More Problems:
    - https://progvar.fun/problemsets/bfs
    - https://progvar.fun/problemsets/grids
    - https://progvar.fun/problemsets/bipartite-graphs
    - codeforces上的一个题单（原版改）
    - Problem Topics

(1-2) Week 8 (Shortest Path: Dijkstra, Floyd-Warshal, Bellman-Ford)

Tutorials:

1. গ্রাফ থিওরিতে হাতেখড়ি-৯ (ডায়াক্সট্রা)
2. গ্রাফ থিওরিতে হাতেখড়ি ১০: ফ্লয়েড ওয়ার্শল

3. গ্রাফ থিওরিতে হাতেখড়ি ১১: বেলম্যান ফোর্ড
4. Bellman-Ford Algorithm - shortest paths with negative weights
5. Dijkstra Algorithm
6. Shortest Paths with Non-Negative Edge Weights (Gold)
7. Video Tutorial with code: https://www.youtube.com/watch?v=CLnpzCnSDSY&list=PL2q4fbVm1Ik64I3VqbVGRfl_OgYzvzt9m&index=7
8. (Tushar Roy): Bellman-Ford Algorithm Single Source Shortest Path Graph Algorithm

Problems:

1. Easy:
   - CSES Problem Set - Tasks (solve Shortest Routes II to Investigation)
   - Cycle Detection
   - Shortest Path
   - Floyd - Warshall + Transitive closure
2. Medium-Hard:
   - Birthday Gift (Dijkstra)
   - Dijkstra/Floyd Warshall_Category(LightOJ)
   - Bellman Ford_Category(LightOJ)
   - https://progvar.fun/problemsets/shortest-paths-dags
   - https://progvar.fun/problemsets/shortest-paths-single-source
   - https://progvar.fun/problemsets/shortest-paths-single-source-negative-weights
   - ProgVar.Fun
   - https://vjudge.net/contest/341347#overview

(1-2) Week 9 (Topological Sort, Tree-Diameter)

Tutorials: (Topological Sort)

1. গ্রাফ থিওরিতে হাতেখড়ি ৭:টপোলোজিকাল সর্ট
2. Topological Sorting
3. টপোলজিকাল সর্ট - smilitude

Problems: (Topological Sort)

1. TOPOSORT - Topological Sorting
2. UVa - 10305 - Ordering Tasks
3. UVa - 124 - Following Orders
5. UVa - 200 - Rare Order
6. Problem - 510C

7. UVa - 11060 - Beverages
8. PFDEP - Project File Dependencies
9. Course Schedule
10. RPLA - Answer the boss!
11. UVa - 452 - Project Scheduling

Tutorials: (Tree Diameter)

1. ট্রি ডায়ামিটার
2. Tree Diameter — Why Does Two BFS Solution Work? | by Tarek Badr

Problems: (Tree Diameter)

1. LightOJ-1094
2. LOJ-1257
3. UVa - 10459
4. 102694A - 3 - Circumference of a Tree
5. Codeforces -102694B

(1-2) Week 10 (Basic Geometry)

Before Starting:

7. Go through and understand every theorem in secondary school (These theorems are must for solving, analyzing geometry problems). Ensure a better understanding and clear concept of vector geometry, straight lines, conics (These topics are covered in intermediate).
8. If you have done 1, go through this book " Handbook of geometry for competitive programmers "

Tutorials:

1. Programming-Problem-In-Bengali/Geometry Resources.md at master · hasancse91/Programming-Problem-In-Bengali
2. Read "Geometry Concepts" tutorials in topcoder Community - Competitive Programming - Tutorials
3. Geometry Algorithms Home
4. Basic Geometry
5. Geometry: 2D points and lines [Tutorial]
6. Geometric Algorithms

Problems:

1. Lightoj (Category - "Basic Geometry")
2. grep+: UVA problems keyword search
3. https://progvar.fun/problemsets/geometry-basics

Level-2 Term-1

Tentative Class Schedule:

Week	Topic
1	Number theory: Extended Euclidean, Euler phi, inverse phi, factorizing n!
2	Graph: SCC, MST
3	Graph: Bicoloring, Bipartite Matching
4	Data Structure: Segment Tree, Lazy
5	Data Structure: BIT, LCA
6	Data Structure: Trie
Online Contest-1	
7	Pattern Matching: KMP, Z-algo
8	DP: LCS, LIS, Matrix-Chain Multiplication
9	DP: Bitmask DP, Digit-DP
10	Geometry: Convex Hull
Online Contest-2	

Guidelines:

- Eat, Sleep, Code, Repeat…

Expectations:

- Total Number of problems solved: 300+
- CF rating:
  -At the start: 1500+
  -At the end: 1800+

Tips:

- By now you must have formed stable teams. You must take part in team contests regularly-at least one every week. Select 2 star contests from CF gym at the beginning, and slowly move on to 3 star contests. Or you may select contests from Timus as well. Discussing unsolved problems after the contest is important too, make sure to solve them. Maintain spreadsheets containing daily logs of what one has solved.

Expected number of problems solved (including previous semester)-800 at the very least. Highly motivated people can go for 1500. You can do it.

Tutorials & Problems List for Level-2 Term-1

(2-1) Week 1 (Number theory: Extended Euclidean, Euler phi, inverse phi, factorizing n!)

Tutorials: (Extended Euclidean)

1. Extended Euclidean Algorithm
2. Extended Euclidean Algorithm

Problems: (Extended Euclidean)

1. UVa – Uva-10104, Uva-10090, Uva-10633, Uva-12775
2. Codeforces Gym – 100963J

Tutorials: (Euler Phi/totient function)

1. Euler's totient function
2. Euler Totient or Phi Function

Problems: (Euler Phi/totient function)

1. SPOJ – ETF, TIP1, ETFS, ETFD, LCMSUM, GCDEX, DCEPCA03, NAJPWG, INVPHI
2. UVa – Uva-10179, 10299, 11327, Uva-10990, 13132, Uva-12995
3. Codechef – COZIE
4. LightOJ – 1007

Tutorials: (Factorizing n!)

1. Prime Factorization of Factorial

Problems: (Factorizing n!)

1. LightOJ – 1035

(2-1) Week 2 (Graph: SCC, MST)

Tutorials: (SCC)

1. গ্রাফ থিওরিতে হাতেখড়ি ১৪ – স্ট্রংলি কানেক্টেড কম্পোনেন্ট (Recommended)
2. Strongly Connected Components and Condensation graph (Recommended)
3. Strongly Connected Components Tutorials & Notes | Algorithms
4. Tarjan's Algorithm to find Strongly Connected Components

Problems: (SCC)

1. Codechef: CHEFRUN
2. LightOJ: 1034, 1003, 1168, 1210, 1406, 1429, 1390, 1417
3. UVA: 247, 11504, 11709, 11770, 11838, 10731, 1229

4. SPOJ: TFRIENDS, CAPCITY, ADAPANEL, GOODA
5. DevSkill: DCP-79
6. Codeforces: 22E

Tutorials: (MST)

1. গ্রাফ থিওরিতে হাতেখড়ি ৬: মিনিমাম স্প্যানিং ট্রি(ক্রুসকাল অ্যালগরিদম) (Kruskal)
2. Minimum spanning tree - Kruskal's algorithm - Competitive Programming Algorithms (Kruskal)
3. Minimum spanning tree - Kruskal with Disjoint Set Union (Kruskal)
4. গ্রাফ থিওরিতে হাতেখড়ি ৫: মিনিমাম স্প্যানিং ট্রি(প্রিম অ্যালগোরিদম) (Prims)
5. Minimum spanning tree - Prim's algorithm (Prims)
6. Second Best Minimum Spanning Tree (Second Best MST)
7. Kirchhoff's Theorem (Kirchhoff's Theorem (To Find number of spanning Tree))

Problems: (MST)

1. SPOJ: MST, ULM09, IITKWPCG
2. UVA: Uva-544, Uva-10034, Uva-11228, Uva-908, 11733, 11733, 10462
3. LightOJ: 1002, 1029, 1041, 1040, 1059, 1123
4. Codeforces: 1095F, 160D, 125E
5. Devskill: DCP-344

(2-1) Week 3 (Graph: Bicoloring, Bipartite Matching)

Tutorials:

1. Check whether a graph is bipartite (Bipartite Checking)
2. Matching Algorithms (Graph Theory)
3. Maximum Bipartite Matching by Kuhn's Algorithm (Maximum Bipartite Matching by Kuhn's Algorithm)
4. I, ME AND MYSELF !!!: Maximum Matching (Hopcroft) (Hopcroft-Karp)
5. গ্রাফ থিওরি: স্টেবল ম্যারেজ প্রবলেম

Problems:

1. UVA: 10004, Uva-11080, Uva-11396, 10505, 10080
2. SPOJ: BUGLIFE
3. LightOJ: 1009, 1149, 1184, 1403, 1201, 1209, 1152, 1304, 1218, 1373, 1206, 1150, 1429, 1242, 1171, 1356
4. Codeforces: 217A, 277A, 862B, 1144F, 1296E1, 664D, 120H, 387D, 468B, 27D, 741C, gym-102006K
5. Atcoder: ABC-131F
6. ProgVar.Fun

(2-1) Week 4 (Data Structure: Segment Tree, Lazy)

Tutorials:

1. Shafayet Blog (Segment Tree): ডাটা স্ট্রাকচার: সেগমেন্ট ট্রি-১
2. Shafayet Blog (Lazy): ডাটা স্ট্রাকচার: সেগমেন্ট ট্রি-২ (লেজি প্রপাগেশন)
3. Codeforces (Dark knight Blog): Algorithm Gym :: Everything About Segment Trees
4. CP-Algorithms: Segment Tree
5. Hacker-Earth: Segment Trees Tutorials & Notes | Data Structures

Problems:

1. SPOJ: MULTQ3, SEGSQRSS, CNTPRIME, KGSS, KQUERY, GSS3
2. Codeforces: 52C, 380C, 295A, 339D, 1234D, 356A, 474F, 920F
3. LightOJ: 1080, 1082, 1083, 1085, 1087, 1089, 1093, 1112, 1135, 1183, 1188, 1207

(2-1) Week 5 (Data Structure: BIT, Sparse Table, LCA)

Tutorials:

1. Shafayet BIT: ডাটা স্ট্রাকচার: বাইনারি ইনডেক্সড ট্রি (Must read)
2. Shafayet LCA: লোয়েস্ট কমন অ্যানসেস্টর (Must read)
3. Hackerearth: Binary Indexed Tree or Fenwick Tree
4. CF gvikei's blog: Basic Binary Indexed Tree (English version)
5. Algorithms Live: Episode 0 - Fenwick Trees
6. Hackerrank (RMQ method): Programming Problems and Competitions
7. Binary Lifting Method: Lowest Common Ancestor - Binary Lifting

Problems:

1. ICPCLive: 6139 - Interval Product, 4806 - Bingo!
2. SPOJ: INVCNT, NICEDAY, ADACABAA, SUMSUM, LCA, DQUERY, DISQUERY, CTRICK, QTREE2
3. LightOJ: 1101, 1112, 1128, 1266, 1348
4. Timus: 1752
5. UVa: 12238, 10938

(2-1) Week 6 (Data Structure: Trie)

Tutorials:

1. ডাটা স্ট্রাকচার: ট্রাই (প্রিফিক্স ট্রি/রেডিক্স ট্রি) (Must read)
2. https://youtu.be/AXjmTQ8LEoI (Must watch)
3. Trie (Keyword Tree) Tutorials & Notes | Data Structures

4. Tutorial on Trie and example problems - Threads @ IIIT Hyderabad
5. Trie | (Insert and Search)
6. https://paste.ubuntu.com/p/qSJN4Ty96P/
7. https://paste.ubuntu.com/p/WCXNrkqXJJ/

Problems:

8. SPOJ: ADAINDEX, PHONELST, SUBXOR, XORX, TRYCOMP, ADAINDEX
9. Codechef: BANKPASS, SUBXOR
10. POJ: 2001
11. UVALive: 4682, 8015
12. UVA: 11488, 12506
13. LightOJ: 1129, 1269, 1114
14. Codeforces: 282E

(2-1) Week 7 (Pattern Matching: KMP, Z-algo)

Tutorials:

2. Shafayet (KMP): স্ট্রিং ম্যাচিং: নুথ-মরিসন-প্র্যাট (কেএমপি) অ্যালগরিদম
3. CP-algorithms (KMP): Prefix function. Knuth–Morris–Pratt algorithm - Competitive Programming Algorithms
4. Hackerearth (Z-algo): Z Algorithm Tutorials & Notes | Algorithms
5. CP-algorithms(Z-algo): Z-function and its calculation
6. CF paladin8's blog: Z Algorithm

Problems:

1. ICPCLive: 6439 - Pasti Pas!
2. LightOJ: 1255, 1258
3. SPOJ: NAJPF
4. Codeforces: 126B, 432D,471D, 808G
5. CodeChef: CDVA1606
6. UVa: 455, 11022, 11452, 11475, 12604, 12467, 11019

(2-1) Week 8 (DP: LCS, LIS, Matrix-Chain Multiplication)

Tutorials:

1. ডাইনামিক প্রোগ্রামিং: লংগেস্ট কমন সাবসিকোয়েন্স [পুরানো ভার্সন]
2. ডাইনামিক প্রোগ্রামিং-৩ (লংগেস্ট ইনক্রিজিং সাবসিকোয়েন্স, পাথ প্রিন্টিং)
3. ডাইনামিক প্রোগ্রামিং ৪ (লংগেস্ট কমন সাবসিকোয়েন্স)
4. ডাইনামিক প্রোগ্রামিং ৭ (ম্যাট্রিক্স চেইন মাল্টিপ্লিকেশন)
5. LightOJ: Longest Increasing Subsequence
6. Longest Increasing Subsequence

7. Matrix Chain Multiplication
8. Matrix Chain Multiplication

Problems:

1. SPOJ: ELIS, LMIS, LCS, NAJLG, ADFRUITS, ROCK, MIXTURES, LISA
2. LightOJ: 1277, 1110, 1157
3. UVa: 481, 231, 10534, 111, 497, 437, 348, 10131, 10066, 12045
4. Codechef: OSQUE, CIRMERGE

(2-1) Week 9 (DP: Bitmask DP, Digit-DP)

Tutorials:

1. বিটমাস্ক ডাইনামিক প্রোগ্রামিং
2. A little bit of classics: dynamic programming over subsets and paths in graphs
3. Shakil Ahmed's Blog : Digit Dp
4. Digit DP
5. Tutorial for Digit Dp - general
6. Programming Problems and Competitions
7. DP Tutorial and Problem List

Problems:

1. SPOJ: GONE, RAONE, CPCRC1C, LUCIFER, ASSIGN
2. CF: 431D, 628D, 215E
3. LightOJ: 1068, 1122, 1125, 1205, 1011, 1057, 1119, 1228
4. Timus: 1152, 1817
5. Codechef: Tools, PPXOR, Chefshop

(2-1) Week 10 (Geometry: Convex Hull)

Tutorials:

1. CP algorithms (Graham's Scan): Convex Hull construction using Graham's Scan
2. Implementation: Convex Hull
3. Hackerrank: Programming Problems and Competitions

Problems:

1. ICPCLive: 3655 - Onion Layers, 4558 - Convex Hull of Lattice Points
2. SPOJ: BSHEEP, VMILI
3. LightOJ: 1203
4. UVA: 10065 - Useless Tile Packers, 681 - Convex Hull Finding
5. Codeforces: 166B
6. Other: Usaco 2014 January Contest, Gold - Cow Curling

Level-2 Term-2

Tentative Class Schedule:

Week	Topic
1	Probability, Expected value
2	Graph: Articulation point, Bridge
3	Graph: FLOW
4	Data Structure: RMQ, BST
5	Data Structure: Heap, SQRT Decomposition
6	Data Structure: HLD
Online Contest-1	
7	DP optimization: Convex Hull Trick
8	DP optimization: Divide and conquer trick
9	Game Theory: Nim, Grundy
10	Geometry: Line Sweep
Online Contest-2	

Guidelines:

- Eat, Sleep, Code, Repeat…

Expectations:

- Total Number of problems solved: 300+
- CF rating:
  -At the start: 1800+
  -At the end: 2100+

Tips:

- Expected number of problems solved-1100 at least, and if you want WF, 2000 minimum. You must have solved around 350 Problems in LightOJ by now, and more than 1000+ on Codeforces
- You must also start solving from CF frequently. Solve as much div2D and div2E as possible-(minimum 300 in total, around 500 if you are serious). In team contests, select 3-4 star contests in CF Gym.

Tutorials & Problems List for Level-2 Term-2

(2-2) Week 1 (Probability, Expected value)

Tutorials:

1. প্রোবাবিলিটি: এক্সপেক্টেড ভ্যালু
2. Mathematical Expectation
3. Sums and Expected Value — part 1 (Part - 1)
4. Sums and Expected Value — part 2

Problems:

1. UVa - 11762, 11427, 11348, 10777
2. LightOJ - Probability/Expected Value Category
3. Codeforces - Probability Tag

(2-2) Week 2 (Graph: Articulation point, Bridge)

Tutorials:

1. গ্রাফ থিওরিতে হাতেখড়ি ১৩: আর্টিকুলেশন পয়েন্ট এবং ব্রিজ
2. Finding articulation points in a graph in O(N+M) - Competitive Programming Algorithms
3. Finding bridges in a graph in O(N+M)
4. Articulation points and bridges (Tarjan's Algorithm)

Problems:

1. UVa - 315, 610, 796, 10199, 10765
2. Codeforces - 700C, 732F, 118E, 1000E
3. LightOJ - Articulation/Bridge Category

(2-2) Week 3 (Graph: FLOW)

Tutorials:

1. গ্রাফ থিওরিতে হাতেখড়ি ১২ – ম্যাক্সিমাম ফ্লো (১)
2. গ্রাফ থিওরিতে হাতেখড়ি-১২ – ম্যাক্সিমাম ফ্লো (২)
3. Maximum flow - Ford-Fulkerson and Edmonds-Karp
4. Maximum flow - Dinic's algorithm
5. Minimum-cost flow - Successive shortest path algorithm
6. Part 1: Key Concepts

Problems:

1. UVa - 11380, 10804, 11506, 10511, 1306, 259

2. CF - 498C, 1288F, 237E, 863F, 316C1, 510E, 311E
3. LightOJ - Max flow-Min cut Category

(2-2) Week 4 (Data Structure: RMQ, BST)

Tutorials: (RMQ)

1. RMQ task (Range Minimum Query - the smallest element in an interval)
2. Sparse Table
3. Also see segment tree, lazy, BIT

Problems: (RMQ)

1. LightOJ – 1082, 1101, 1093, 1083, 1082, 1112, 1080, 1164, 1087, 1183, 1135
2. SPOJ - RMQSQ
3. Codechef - FRMQ

Tutorials: (BST)

1. Binary Search Tree(BST)
2. Binary Search Tree
3. Data Structure - Binary Search Tree

Problems: (BST)

1. Hackerrank: Binary Search Tree Problems
2. Misc

(2-2) Week 5 (Data Structure: SQRT Decomposition, Heap)

Tutorials: (SQRT Decomposition)

1. স্কয়ার-রুট ডিকম্পোজিশন
2. Square Root Decomposition | Tanvir's Blog
3. Learn Mo's Algorithm and Mo's algorithm on Tree (with practice problems)
4. SQRT decomposition
5. Everything on Mo's Algorithm
6. Mo's Algorithm on Trees [Tutorial]
7. Square Root Decomposition (Platinum)

Problems: (SQRT Decomposition)

1. Codeforces: 86D, 786C, 840D, 13E, 617E, 375D, 342E, 44C, 455D, 398D, 220B,446C, 487D, 506D, 348C, 444C.
2. LOJ: 1082, 1093,
3. Codechef: GERALD07, Sherlock and Inversions (IITI15),VLB,

4. SPOJ: COT2, DQUERY, GIVEAWAY
5. Uva: 12003, 11990

Tutorials: (Heap)

1. Heaps/Priority Queues Tutorials & Notes | Data Structures
2. Heap Data Structure
3. Heap Data Structure

Problems: (Heap)

1. Codeforces: Problems list in a blog
2. Hackerrank: Heap/Priority Queue Problems

(2-2) Week 6 (Data Structure: HLD)

Tutorials:

1. Anudeep's Blog (Must read): Heavy Light Decomposition (Note: This site may show warning! Go to -> details -> proceed to this site)
2. Geeksforgeeks: Heavy Light Decomposition | Set 1 (Introduction)
3. E-maxx: Heavy-light decomposition
4. Curious to learn more?: (Have a look at this CF blog) Easiest HLD with subtree queries
5. Also this: Hybrid Tutorial #-1: Heavy-Light Decomposition

Problems:

1. See the problems from this contest:
   - Link: https://vjudge.net/contest/325844
   - Pass: too_heavy

(2-2) Week 7 (DP optimization: Convex Hull Trick)

Tutorials:

1. Dynamic Programming Optimization - Convex Hull Trick (Must read)
2. Convex hull trick and Li Chao tree
3. Convex Hull Trick
4. [Tutorial] Convex Hull Trick — Geometry being useful
5. Episode 11 - Convex Hull Optimization

Problems:

1. See the problems from this contest:
   - Link: https://vjudge.net/contest/391342 (See Problems: F - P)
   - Pass: dp_optimize

(2-2) Week 8 (DP optimization: Divide and conquer trick)

Tutorials:

1. What is divide and conquer optimization in dynamic programming? (Quora)
2. Divide and Conquer DP (CP algorithms)
3. Divide and Conquer Optimization (jeffreyxiao)
4. Divide and Conquer Optimization in Dynamic Programming (opengenus)

Problems:

1. See the problems from this contest:
    - Link: https://vjudge.net/contest/391342 (See Problems: A - E)
    - Pass: dp_optimize

(2-2) Week 9 (Game Theory: Nim, Grundy)

Tutorials:

1. Shafaeter Blog: গেম থিওরি - ১
2. Nim Game Example:
    - Nim Game Example
    - Algorithm Games
3. Staircase Nim: Intro to Staircase Nim + Editorial for HackerRank "Move the Coins"
4. Grundy:
    - Sprague-Grundy theorem. Nim
    - Grundy numbers for competitive programming
5. Graph: Games on arbitrary graphs
6. Others:
    - https://www.math.ucla.edu/~tom/Game_Theory/comb.pdf
    - Tutorial for A Coin Game
    - Combinatorial games
    - Theory of Impartial Games
    - Combinatorial Game Theory

Problems:

1. See the problems from the following contest:
    - Link: https://vjudge.net/contest/321886
    - Pass: lets_play_a_game

(2-2) Week 10 (Geometry: Line Sweep)

Tutorials:

1. Line Sweep Algorithms
2. Line Sweep Technique Tutorials & Notes | Math
3. How to sweep like a Sir
4. Search for a pair of intersecting segments
5. Point location in O(log n)

Problems:

1. LightOJ: 1120
2. TopCoder: BoxUnion, CultureGrowth, PowerSupply, ConvexPolygons
3. UVa: 12657, 12647, 12310
4. Timus: 1848, 1469
5. A2OJ: Line Sweep Problems
6. Codechef: Line Sweep Tag

Level-3 + Level-4

Guidelines:

- Eat, Sleep, Code, Repeat…
- Learn on your own
- Take care of your juniors (Take classes, monitor their performance, motivate and help them)

Tips:

- By now you must have learnt the basic and intermediate things of almost all topics. Start specializing, discussing among teammates.
- Try to overcome your weak points and learn the things that you missed before. Focus on more advanced topics.
- For example, you decide to be more specialized in DP. Learn complex DP techniques like Knuth Optimization, etc.
- Or if you want to specialize in Math, learn FFT, NTT and more advanced number theory topics.
- For data structure, splay tree, treap, HLD(solve QTREEs from SPOJ) etc.
- For graph, more esoteric problems.

I mean, by now you must have a good idea of what topics you want to do, and further specialize on them. The above are the topics that must be learnt at least by one member of the team, but there may be rarer topics as well.

Expectation: (For the rest of your Contest Career)

- Achieve top ranks in onsite contests
  - -Top in divisional
  - -Top-5 in national
- Assuming you have worked hard, you have become the best contestant CUET has ever seen. Your name and fame spreads in BD competitive programming landscape.
- **Qualify for the World Finals ☺**